

BUBT Bus Tracking System

Actor Authentication & Credentials Specification

This document defines how each of the three actors (User, Driver, Administrator) registers, logs in, and is provisioned in the system. It also includes the fixed seed Administrator credentials for initial setup.

1. User (Student / Teacher / Staff)

Registration Flow

- User signs up with Name, Email, Password, and optional Designation (Student / Teacher / Staff) + ID Number.
- Account is created with status = **unverified** and cannot log in yet.
- System sends a 6-digit OTP code to the provided email (valid for a limited time, e.g. 10 minutes).
- User enters the OTP on the verification screen → account status becomes **verified** → login is now allowed.
- Unverified accounts attempting login are redirected to the OTP screen with a “Resend OTP” option (rate-limited to prevent spam, e.g. one resend per 60 seconds).
- OTP is single-use and invalidated after successful verification or expiry, whichever comes first.

Registration Fields

Field	Required?	Notes
Full Name	Required	-
Email	Required	Used for login + verification link
Password	Required	Hashed with bcrypt, min. 8 characters
Designation	Optional	Student / Teacher / Staff — for display only, no permission difference
ID Number	Optional	University Student/Employee ID, for display only

Login

Email + Password → JWT access token (short-lived) + refresh token issued on success. Login is blocked until the OTP verification step above is completed.

2. Driver

Drivers never self-register. An Administrator creates every driver profile from the Admin Panel. The system auto-generates a Driver ID and a temporary password at creation time, which the Admin shares with the driver.

Driver Profile Fields (set by Admin)

Field	Required?	Notes
Driver ID	Auto-generated	e.g. DRV-001, used as the login username
Full Name	Required	-

Phone Number	Required	Used for the in-app “Call Driver” feature
Blood Group	Required	For emergency purposes
Email	Optional	For notifications only, not used to log in
Photo	Optional	Shown to users tracking the trip
NID / License Number	Required	Internal record-keeping
Address	Required	Internal record-keeping
Emergency Contact	Required	A second contact number
Assigned Bus	Required	One driver → one bus, dropdown of existing buses
Joining Date	Auto (editable)	Defaults to today
Status	Required	Active / Inactive toggle — inactive drivers can’t log in
Temporary Password	Auto-generated	Shown once to Admin at creation; driver should change it on first login

Login

Driver ID + Password → JWT access token. First login prompts a forced password change.

Password Reset

No self-service reset. If a driver forgets their password, an Admin resets it from the Admin Panel and shares the new temporary password directly.

3. Administrator

Administrators are not created through any UI in v1 — there is no “add admin” screen. Two fixed seed accounts are created directly in the database at setup time.

Seed Admin Accounts

■ Development / Seed Credentials

Admin ID (username)	Password	Purpose
admin.transport1	Bubt@Trans2026!A	Primary Administrator
admin.transport2	Bubt@Trans2026!B	Secondary / backup Administrator

These two accounts are inserted via the database seed script (Phase 3/6) so the system is usable immediately after deployment, before any real admin accounts exist. **These exact passwords must be changed immediately after first login in any real deployment** — they are placeholders for development/demo purposes only and should never be used in production as-is.

Login

Admin ID + Password → JWT access token, same auth endpoint as Users/Drivers, differentiated by the **role** field in the shared users table.

4. Summary: Who Can Log In, and How

Actor	Can Self-Register?	Login Identifier	Provisioned By
User	Yes (email verification required)	Email	Self
Driver	No	Driver ID	Administrator
Administrator	No	Admin ID	Fixed DB seed (v1)

5. Note on Supplied UI Screenshot

A UI screenshot was provided as a visual/color reference only (primary blue #2563EB-ish, white cards, rounded corners, soft shadows, green status badges). Its content (e.g. “Your Assigned Bus”) does **not** reflect actual business logic — confirmed that users are not assigned to a fixed bus. The color palette will be formally adopted in Phase 5 (UI/UX Planning).